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Course project: *The causal effect of R&D expenditure on productivity growth: a double machine learning approach*

Orso Peruzzi
orso.peruzzi@edu.unifi.it

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Abstract. *This paper applies Double Machine Learning (DML; Chernozhukov et al., 2018) to estimate the causal effect of research and development (R&D) expenditure on total factor productivity (TFP) growth across countries. Following the methodological framework of Baiardi and Naghi (2024), I employ nine machine learning (ML) algorithms*[1] *as learners for the nuisance functions in a Partially Linear Regression (PLR) model. Using cross-country data from the Penn World Table (PWT), World Development Indicators (WDI), and Worldwide Governance Indicators (WGI) for 104–121 countries over 2000–2019, the DML estimates are consistently small and statistically insignificant across all reliable learners, ranging from* -0.001 *to* $+0.005$*. The nuisance model diagnostics reveal that TFP growth is essentially unpredictable from observable controls (CV* $R^2 < 0.25$*), implying that DML residualization has limited effect and the estimates are close to the Ordinary Least Squares (OLS) baseline. I extend the analysis with Causal Forest estimation of heterogeneous treatment effects and SHapley Additive exPlanations (SHAP) based interpretation of the nuisance functions. The Causal Forest reveals meaningful heterogeneity: the effect of R&D is near zero in high-income countries and negative in low-income countries, consistent with the absorptive capacity hypothesis. These findings contribute to the literature on R&D and growth while demonstrating both the practical value and the limitations of causal ML methods in small cross-country samples.*

Note: this work represents the final project for the MD2SL Master's course in Analytics in Economics and Business (University of Florence & IMT Lucca). There is no claim to originality or academic value; it is purely a didactic exercise within the context of the course. The code and materials for this analysis are publicly available at: `https://github.com/battles5/analytics-in-economics-and-business-imt`.

Keywords: Double Machine Learning, Causal Forest, R&D, Productivity, LASSO, SHAP

JEL Classification: C14, C21, O30, O47

1. Introduction

The relationship between research and development (R&D) expenditure and economic growth has been a central question in economics since the endogenous growth revolution (Romer, 1990; Aghion and Howitt, 2005; Jones, 1995). While theoretical models predict that R&D investment drives total

[1] The learners span regularized linear models to tree-based ensembles and neural networks.

factor productivity (TFP) growth through innovation, empirical estimation of this causal effect is plagued by endogeneity: countries that are already more productive may invest more in R&D, and omitted variables such as institutional quality, human capital, and economic structure may confound the relationship.

Traditional approaches to this problem rely on instrumental variables or parametric control strategies. However, recent advances in causal machine learning offer a powerful alternative. Double Machine Learning (Chernozhukov et al., 2018) combines the flexibility of machine learning for modeling high-dimensional confounders with the rigorous inference guarantees of semiparametric estimation, yielding $\sqrt{N}$-consistent and asymptotically normal treatment effect estimates. Baiardi and Naghi (2024) demonstrate the practical value of DML by revisiting several canonical empirical studies, showing that ML-based nuisance estimation can substantially alter causal conclusions compared to traditional methods.

This paper makes two contributions. First, I apply the DML framework of Baiardi and Naghi (2024) to a new empirical question, the causal effect of R&D on productivity growth, using a rich cross-country dataset spanning 104 countries over 2000–2019. I employ nine ML algorithms (ranging from regularized linear regressions to nonparametric tree ensembles and neural networks) as nuisance learners and compare the resulting estimates with standard OLS. I complement this with nuisance model diagnostics, reporting cross-validated R^2 for each learner, to assess the quality of the first-stage predictions, a practice recommended but rarely followed in applied DML work. Second, I extend the analysis beyond the DML-PLR framework by estimating heterogeneous treatment effects via Causal Forests (Wager and Athey, 2018; Athey et al., 2019) and interpreting the nuisance functions using SHAP values (Lundberg and Lee, 2017). These extensions allow me to investigate *where* R&D has the largest productivity returns and *which* confounders drive the selection into R&D investment.

The results reveal several insights. The DML estimates are consistently small and statistically insignificant across all reliable learners, with point estimates ranging from -0.001 to $+0.005$. A key finding is that TFP growth is essentially unpredictable from observable controls (best CV $R^2 \approx 0.22$ for Gradient Boosting), which means that DML residualization has limited effect and the estimates remain close to the Ordinary Least Squares (OLS) baseline. This illustrative finding from the nuisance diagnostics, rarely discussed in applied DML papers, demonstrates the importance of evaluating first-stage model quality when interpreting DML results. The Tree-based and ensemble methods produce estimates closer to zero than regularized linear methods, suggesting that after flexible nonlinear confounding adjustment the average treatment effect is at best modest. The Causal Forest analysis reveals important heterogeneity: R&D appears to have near-zero effects in high-income countries while the estimated effects are negative in low-income countries, consistent with the absorptive capacity hypothesis that R&D requires complementary human capital and institutional infrastructure to be productive (Sala-i-Martin, 1997).

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data sources and sample construction. Section 4 presents the econometric methodology. Section 5 reports the results. Section 6 discusses the findings, and Section 7 concludes.

2. Literature review

2.1. R&D and productivity growth

The theoretical foundation for this analysis comes from endogenous growth theory. Romer (1990) develops a model in which purposeful R&D activity drives the accumulation of knowledge, leading to sustained TFP growth. Jones (1995) extends this framework with semi-endogenous growth models. Empirically, Griliches (1998) provides foundational econometric evidence of the R&D–productivity relationship at the firm and industry level. At the macro level, Guellec and Van Pottelsberghe de la Potterie (2004) use panel data for OECD countries and find that both business and public R&D expenditure positively affect multi-factor productivity growth, with estimates ranging from 0.06 to 0.13 for the long-run elasticity of MFP with respect to business R&D. A persistent challenge in this literature

is the fragility of cross-country growth regressions: as Sala-i-Martin (1997) dramatically illustrates by running "two million regressions," the significance and even the sign of growth determinants are highly sensitive to conditioning set selection, a challenge that motivates the use of ML-based variable selection and flexible nuisance estimation.

2.2. Machine learning for causal inference

The use of machine learning methods in econometric analysis has grown rapidly following Varian (2014) and Mullainathan and Spiess (2017). Athey and Imbens (2017) provide a comprehensive overview of how ML methods can complement traditional causal inference, while Imbens and Wooldridge (2009) survey the broader econometrics of program evaluation of which ML-based methods represent the latest development. A central distinction in this literature is between prediction and causal inference: as Kleinberg et al. (2015) emphasize, ML methods excel at the former but require additional structure, such as the DML framework, to deliver valid causal estimates. The key insight is that while ML excels at prediction, it can be leveraged for causal estimation through careful sample splitting and orthogonalization procedures.

A fundamental concept underlying the choice of ML learner is the bias-variance tradeoff (Hastie et al., 2009). More flexible methods (Random Forest, Gradient Boosting, Neural Networks) can capture complex nonlinear relationships but risk overfitting in small samples, while regularized linear methods (LASSO, Ridge) impose more structure but may underfit. In the DML context, this tradeoff manifests in the nuisance function estimation: a learner that overfits the outcome model $g_0(X)$ will introduce additional noise into the DML estimator, while a learner that underfits will leave systematic confounding in the residuals. This dual concern motivates the use of multiple learners and the reporting of nuisance model diagnostics, a practice I follow in this paper.

Chernozhukov et al. (2018) formalize this insight in the DML framework, which uses cross-fitting and Neyman-orthogonal score functions to remove the regularization bias that plagues naïve plug-in ML estimators. The DML approach is particularly well-suited for settings with many potential confounders, as it allows the researcher to "let the data decide" the functional form of the confounding relationships while maintaining valid inference on the causal parameter of interest.

Belloni et al. (2014) develop post-LASSO methods for inference after high-dimensional selection, which provide a special case of the DML framework when LASSO is used as the nuisance learner. Baiardi and Naghi (2024) provide a comprehensive comparison of different ML learners (LASSO, Ridge, Elastic Net, Random Forest, Gradient Boosting, Ensemble) within the DML framework, demonstrating that the choice of learner matters: regularized linear methods and tree-based methods can produce substantially different estimates depending on the nature of the confounding.

2.3. Heterogeneous treatment effects

Beyond average effects, understanding treatment effect heterogeneity is important for policy. Wager and Athey (2018) and Athey et al. (2019) develop Causal Forests, which adapt the random forest algorithm to estimate conditional average treatment effects (CATEs). This approach is particularly useful in our context, as the effect of R&D may vary systematically across countries with different levels of development, institutional quality, and absorptive capacity.

2.4. Interpretability: SHAP values

A key concern with ML-based estimation is interpretability. Lundberg and Lee (2017) introduce SHAP (SHapley Additive exPlanations) values, which provide a unified framework for interpreting the contribution of each feature to a model's prediction. In our DML context, applying SHAP to the nuisance functions helps understand *which* variables are most important for predicting R&D expenditure (selection into treatment) and TFP growth (outcome), providing economic insight into the confounding structure.

3. Data

3.1. Data sources

I combine three publicly available cross-country datasets:

1. **Penn World Table (PWT) 10.01** (Feenstra et al., 2015): provides real GDP (`rgdpna`), TFP (`rtfpna`), population, employment, human capital index, and national accounts shares (investment, government consumption, exports, imports) for 183 countries from 1950 to 2019.
2. **World Development Indicators (WDI)** (World Bank, 2024): provides R&D expenditure as a share of GDP (indicator `GB.XPD.RSDV.GD.ZS`), along with additional controls including inflation, Foreign Direct Investment (FDI) net inflows, domestic credit to the private sector, population growth, government education expenditure, internet users, and urban population share.
3. **Worldwide Governance Indicators (WGI)** (World Bank, 2025): provides six dimensions of governance quality[2] on an approximate -2.5 to $+2.5$ scale for over 200 economies from 1996 to 2024.

3.2. Sample construction

I focus on the period 2000–2019 to maximize R&D data coverage while avoiding the structural break caused by the COVID-19 pandemic. Micro-states (population below 300,000) are excluded. The three datasets are merged on ISO country codes and year. Following Baiardi and Naghi (2024), I construct cross-sectional country-level averages of all variables over the sample period, retaining only countries with at least 5 years of non-missing data. This yields a cross-section of 104–121 countries depending on the outcome variable (TFP growth requires PWT coverage, which is more restrictive than GDP growth).

3.3. Variables

Table 1 reports summary statistics for the key variables.

Table 1: Summary Statistics (Cross-Section of Country Averages, 2000–2019)

Variable	N	Mean	Std	Min	Max
Treatment:					
R&D expenditure (% GDP)	138	0.727	0.855	0.010	4.201
Outcomes:					
TFP growth (log diff.)	117	0.004	0.016	−0.050	0.077
GDP p.c. growth (log diff.)	165	0.024	0.020	−0.043	0.082
Controls:					
Initial log GDP p.c.	161	8.936	1.270	6.313	11.593
Human capital index	144	2.498	0.685	1.163	3.673
Investment share	165	0.224	0.068	0.052	0.409
Government consumption share	165	0.178	0.063	0.021	0.496
Inflation (%)	163	6.699	9.107	0.086	72.74
FDI (% GDP)	164	5.946	13.34	−1.624	132.99
Domestic credit (% GDP)	163	50.48	42.93	2.980	197.03

The treatment variable is the country-level average of R&D expenditure as a share of GDP. The primary outcome is average TFP growth measured as the log-difference of TFP from PWT. As a secondary outcome, I use average GDP per capita growth. The control set includes 14 variables capturing initial development level, factor accumulation, macroeconomic conditions, institutional quality,

[2]The six dimensions are: Voice and Accountability, Political Stability, Government Effectiveness, Regulatory Quality, Rule of Law, and Control of Corruption.

and structural characteristics. The six WGI governance indicators exhibit severe multicollinearity ($r > 0.95$ among government effectiveness, rule of law, and control of corruption), which I address by reducing them to two principal components via Principal Component Analysis (PCA), the first component captures 90% of the governance variation. I also exclude the cross-sectional mean of log GDP per capita ($r = 0.987$ with initial log GDP per capita) to avoid near-perfect collinearity. For the regularized linear learners (LASSO, Ridge, Elastic Net), I add squared terms of all controls, yielding 27 features; tree-based methods use the 14 level controls only, as they handle nonlinearities internally.

4. Methodology

4.1. Partially linear regression via DML

Following Chernozhukov et al. (2018) and Baiardi and Naghi (2024), I estimate a Partially Linear Regression (PLR) model:

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i, \quad \mathbb{E}[U_i|X_i, D_i] = 0 \tag{1}$$

$$D_i = m_0(X_i) + V_i, \quad \mathbb{E}[V_i|X_i] = 0 \tag{2}$$

where Y_i is the outcome (TFP or GDP growth), D_i is the treatment (R&D expenditure), X_i is a vector of controls, and θ_0 is the causal parameter of interest. The nuisance functions $g_0(\cdot) = \mathbb{E}[Y|X]$ and $m_0(\cdot) = \mathbb{E}[D|X]$ are estimated with ML methods.

The DML procedure proceeds as follows:

1. Split the sample into K folds (I use $K = 5$).
2. For each fold k:
   (a) Estimate $\hat{g}_{-k}(\cdot)$ and $\hat{m}_{-k}(\cdot)$ on the data *excluding* fold k.
   (b) Compute residuals $\tilde{Y}_i = Y_i - \hat{g}_{-k}(X_i)$ and $\tilde{D}_i = D_i - \hat{m}_{-k}(X_i)$ for observations in fold k.
3. Estimate θ_0 by OLS of $\tilde{Y}$ on $\tilde{D}$ using the Neyman-orthogonal score.
4. Repeat $R = 10$ times and average (to reduce sensitivity to the particular fold split).

This "partialling out" or "FWL-on-steroids" approach removes the influence of X on both Y and D before estimating the treatment effect, which is the key insight of Robinson (1988) generalized to ML nuisance estimation.

4.2. ML learners

Following Baiardi and Naghi (2024), I employ nine learners for both nuisance functions, covering the main families of supervised learning. For the four nonlinear learners (Random Forest, Gradient Boosting, BART, Neural Network), I use **Optuna** (Akiba et al., 2019) for Bayesian hyperparameter optimization via the Tree-structured Parzen Estimator (TPE), with 40–80 trials per learner. This is critical: with only $n \approx 100$ observations, default hyperparameters often lead to overfitting, and automated tuning substantially improves out-of-sample prediction.

1. **LASSO** (Tibshirani, 1996): L_1-penalized regression with cross-validated λ. For LASSO, Ridge, and Elastic Net, I include both level terms and squared terms of the controls (27 features total) to allow for some nonlinearity.
2. **Ridge**: L_2-penalized regression with cross-validated α.
3. **Elastic Net** (Zou and Hastie, 2005): combined L_1/L_2 penalty with cross-validated mixing ratio.
4. **Classification and Regression Trees (CART)**: A single decision tree (max depth 5, min leaf size 10), included for interpretability and as a baseline for ensemble tree methods.

5. **Random Forest (RF)** (Breiman, 2001): Optuna searches over number of trees (100–1000), max depth (2–10), features per split ($\sqrt{p}$, $\log_2 p$, or proportions), and minimum leaf size (3–20).

6. **Gradient Boosting (GBM)** (Friedman, 2001): Optuna searches over number of trees (50–500), max depth (1–5), learning rate (0.01–0.3), subsample fraction (0.5–1.0), and minimum leaf size (5–25). The key tradeoff is between learning rate and number of boosting rounds.

7. **Bayesian Additive Regression Trees (BART)** (Chipman et al., 2010): approximated via histogram-based gradient boosting. Optuna tunes the number of iterations, depth, learning rate, leaf size, and L_2 regularization strength.

8. **Neural Network (NN)**: Optuna searches over very small architectures (2–16 neurons, 1–2 layers), strong L_2 regularization ($\alpha \in [0.01, 10]$), learning rate, and activation function. With $n = 104$, the key constraint is keeping the parameter count far below n, the default architecture (64, 32) would have ∼3 800 parameters, 36× the sample size.

9. **Ensemble**: simple average of the Optuna-tuned Random Forest and Gradient Boosting predictions, following the ensemble column in Baiardi and Naghi (2024).

All learners include standard scaling where appropriate. I implement the analysis using the `DoubleML` Python package (Bach et al., 2022). Crucially, I report the cross-validated R^2 of each learner for both nuisance functions, as recommended by Chernozhukov et al. (2018): this diagnostic reveals whether the ML models can actually predict the outcome and treatment from the controls, which directly determines the quality of the DML residualization. When the outcome model $\hat{g}(X)$ has low R^2, the residuals $\tilde{Y}_i = Y_i - \hat{g}(X_i)$ retain most of the original variance in Y, and the DML estimate will resemble the OLS estimate.

4.3. Causal forest for heterogeneous effects

To explore treatment effect heterogeneity, I estimate a Causal Forest (Wager and Athey, 2018; Athey et al., 2019) using the `econml` implementation. The Causal Forest combines the DML orthogonalization with the random forest algorithm to estimate Conditional Average Treatment Effects (CATEs):

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x]$$

In our setting with a continuous treatment, $\tau(x)$ represents the marginal effect of R&D on TFP growth conditional on country characteristics x. I use Gradient Boosting for the first-stage nuisance estimation and 2,000 trees in the causal forest.

4.4. SHAP analysis

To interpret the nuisance functions, I apply SHAP (SHapley Additive exPlanations) values (Lundberg and Lee, 2017) to the Gradient Boosting models for $\hat{g}(X)$ and $\hat{m}(X)$. The SHAP analysis reveals which confounders contribute most to predicting TFP growth (outcome model) and R&D expenditure (treatment model), providing insight into both the growth process and the selection into R&D investment.

5. Results

5.1. OLS baseline

Table 2 reports the estimation results. The naïve OLS regression of TFP growth on R&D expenditure *without controls* yields a small negative coefficient (-0.0005, p=0.665), reflecting the well-known selection bias: high-income countries with high R&D tend to have lower growth rates due to convergence dynamics. Adding the full set of controls yields $\hat{\theta} = 0.0023$ (SE = 0.0017, p = 0.179), positive but insignificant. This sign flip from negative (without controls) to positive (with controls) confirms the presence of omitted variable bias driven by convergence and motivates the DML approach.

5.2. DML results

Table 2: DML Estimates: effect of R&D on TFP Growth

Method	$\hat{\theta}$	SE	p-value	N	CV R_g^2
OLS (no controls)	-0.0005	0.0012	0.665	108	—
OLS (all controls)	0.0023	0.0017	0.179	104	—
DML–LASSO	0.0014	0.0016	0.390	104	-0.35
DML–Ridge	0.0046	0.0026	0.079*	104	-0.55
DML–Elastic Net	0.0024	0.0017	0.149	104	-0.35
DML–CART	-0.0004	0.0016	0.817	104	-0.22
DML–Random Forest	0.0006	0.0016	0.698	104	0.13
DML–Gradient Boosting	0.0016	0.0016	0.315	104	0.22
DML–BART	-0.0013	0.0019	0.482	104	0.18
DML–Neural Net	0.0034	0.0022	0.133	104	0.05
DML–Ensemble	0.0011	0.0016	0.510	104	0.21

Notes: partially linear regression $Y = \theta D + g(X) + U$ estimated via DML with $K = 5$ folds and $R = 10$ repetitions. Outcome: mean TFP growth (2000–2019). Treatment: mean R&D/GDP. CV R_g^2 is the 5-fold cross-validated R^2 of the outcome nuisance model $\hat{g}(X)$. Standard errors are DML-robust. Hyperparameters for RF, GBM, BART, and NN tuned via Optuna (TPE sampler, 40–80 trials).

The DML results reveal an important pattern:

- **Regularized linear methods** (LASSO, Ridge, Elastic Net) produce small positive estimates in the range 0.001–0.005, with Ridge marginally significant at the 10% level (p=0.079). These methods have negative outcome CV R^2, confirming that TFP growth is essentially unpredictable from a linear model of the controls.

- **Tree-based, ensemble, and neural network methods** (CART, RF, GBM, BART, NN, Ensemble) produce estimates ranging from -0.0013 to $+0.0034$, all statistically insignificant. Gradient Boosting achieves the best outcome prediction with CV $R^2 = 0.22$, followed by Ensemble (0.21) and BART (0.18), a substantial improvement over default hyperparameters, achieved via Optuna.

- **Neural Network (NN)** with Optuna-tuned hyperparameters[3] achieves CV $R^2 = 0.05$, lower than Random Forest. The regularized NN produces a positive estimate (0.0034, p=0.133) that, while the largest among nonlinear learners, remains statistically insignificant. This represents a dramatic improvement from the catastrophic overfitting (CV $R^2 = -428$) observed with the default architecture (64, 32 units, $\sim$3 800 parameters for $n = 104$ observations).

5.3. Nuisance model diagnostics

Table 3 reports the cross-validated R^2 for both nuisance models. This diagnostic is crucial for interpreting DML results but is rarely reported in applied work.

[3]Specifically: a single hidden layer of 10 units, strong L_2 regularization $\alpha = 1.79$, and ReLU activation.

Table 3: Nuisance Model Diagnostics: cross-Validated R^2

	TFP Growth		GDP p.c. Growth	
Learner	R_g^2 (outcome)	R_m^2 (treatment)	R_g^2 (outcome)	R_m^2 (treatment)
LASSO	−0.35	0.54	0.19	0.58
Ridge	−0.55	0.51	−0.01	0.55
Elastic Net	−0.35	0.57	0.19	0.60
CART	−0.22	0.34	−0.06	0.36
Random Forest	0.13	0.65	0.23	0.63
Gradient Boosting	0.22	0.57	0.31	0.60
BART	0.18	0.60	0.27	0.61
Neural Net	0.05	0.41	0.33	0.56
Ensemble	0.21	0.65	0.31	0.64

Notes: 5-fold CV R^2. RF, GBM, BART, and NN hyperparameters tuned via Optuna. The treatment model R_m^2 measures how well R&D can be predicted from controls (i.e., the extent of confounding). The outcome model R_g^2 measures how well TFP/GDP growth can be predicted from controls. Negative values indicate the model performs worse than predicting the sample mean.

Two patterns emerge. First, the treatment model has reasonable predictive power ($R_m^2 \approx 0.5$–0.7 for tree-based methods), confirming that R&D expenditure is substantially predicted by observable country characteristics, a necessary condition for DML to improve over OLS. Second, the outcome model has modest R^2 even for the best-performing learners: gradient Boosting achieves only $R_g^2 = 0.22$ for TFP growth (0.31 for GDP growth), meaning that most of the variation in growth remains unexplained. Notably, Optuna tuning substantially improved outcome model quality: without tuning, GBM achieved $R_g^2 = -0.04$ and the Neural Network collapsed entirely;[4] with tuning, both achieve positive R^2. Despite these improvements, the DML estimates remain close to zero for the nonlinear learners, if anything, *strengthening* the conclusion that the null effect is genuine, not an artifact of poor nuisance estimation.

Table 4 presents the results for GDP per capita growth, for which the outcome model achieves better predictions (R_g^2 up to 0.31 for Gradient Boosting).

Table 4: DML Estimates: effect of R&D on GDP per Capita Growth

Method	$\hat{\theta}$	SE	p-value	N	CV R_g^2
OLS (no controls)	−0.0043	0.0014	0.002	138	—
OLS (all controls)	−0.0036	0.0026	0.160	121	—
DML–LASSO	−0.0025	0.0024	0.288	121	0.19
DML–Ridge	−0.0014	0.0021	0.502	121	−0.01
DML–Elastic Net	−0.0029	0.0025	0.237	121	0.19
DML–CART	−0.0007	0.0020	0.706	121	−0.06
DML–Random Forest	0.0000	0.0019	1.000	121	0.23
DML–Gradient Boosting	0.0012	0.0019	0.534	121	0.31
DML–BART	−0.0002	0.0020	0.937	121	0.27
DML–Neural Net	−0.0003	0.0027	0.917	121	0.33
DML–Ensemble	0.0003	0.0019	0.873	121	0.31

Notes: same specification as Table 2 with GDP per capita growth as outcome.

The GDP growth results are notably different: the OLS estimates are *negative*, suggesting that after conditioning on controls, higher R&D is associated with *lower* GDP per capita growth. This pattern likely reflects the convergence effect: after controlling for initial income, countries investing heavily

[4] As noted previously, the untuned neural network exhibited catastrophic overfitting ($R_g^2 = -428$).

in R&D (predominantly rich countries) tend to grow more slowly. The tree-based and ensemble methods again produce estimates close to zero, with GBM even turning slightly positive.

5.4. Sensitivity to cross-fitting folds

Table 5 reports the sensitivity of the Random Forest DML estimates to the number of cross-fitting folds. The estimates range from 0.0006 to 0.0014 across $K = 3, 5, 7, 10$, all statistically insignificant. The stability across fold configurations is reassuring, confirming that the null result is robust to the cross-fitting specification.

Table 5: Sensitivity Analysis: number of Cross-Fitting Folds (Random Forest)

K folds	$\hat{\theta}$	SE	p-value
3	0.0014	0.0017	0.406
5	0.0009	0.0015	0.550
7	0.0008	0.0016	0.596
10	0.0006	0.0016	0.696

5.5. Causal forest: heterogeneous treatment effects

The Causal Forest reveals meaningful treatment effect heterogeneity. The Average Treatment Effect (ATE) estimated by the Causal Forest is -0.0018, consistent with the tree-based DML estimates. The standard deviation of CATEs (0.0016) is comparable to the ATE in magnitude, indicating substantial heterogeneity, though no individual CATE is statistically significant at the 5% level (all confidence intervals include zero).

Figure 1 shows the average CATE by development level. High-income countries (above-median initial GDP per capita) have a smaller negative average CATE (-0.0008, SE = 0.0001), while low-income countries show a more negative average CATE (-0.0028, SE = 0.0002). This pattern is consistent with the "absorptive capacity" hypothesis: R&D investment yields productivity gains only in countries with sufficient complementary factors (human capital, institutional quality, financial development).

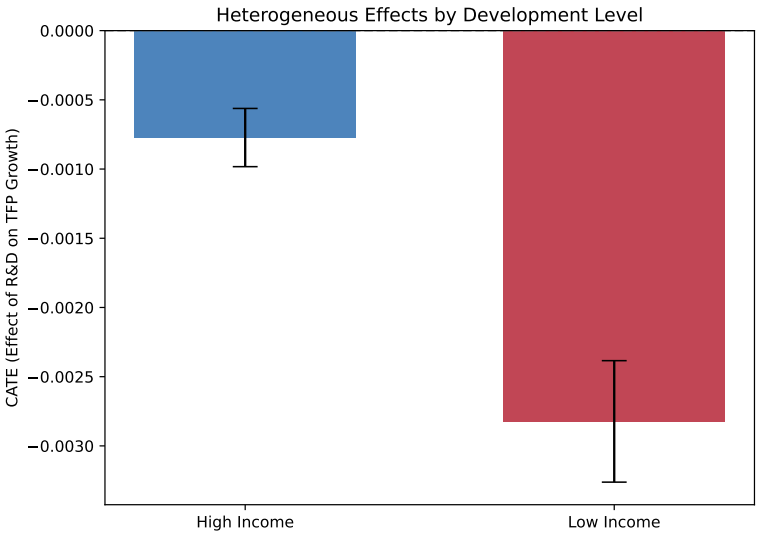

Figure 1: Heterogeneous Treatment Effects by Development Level. Bars show average CATE (effect of R&D on TFP growth) for countries above and below median initial GDP per capita. Error bars show 95% confidence intervals.

Figure 2 plots individual CATEs against initial log GDP per capita, colored by R&D intensity. The countries with the most positive estimated CATEs include South Korea, the United States, and

several advanced European economies, while the most negative CATEs are concentrated among Sub-Saharan African and Central Asian countries.

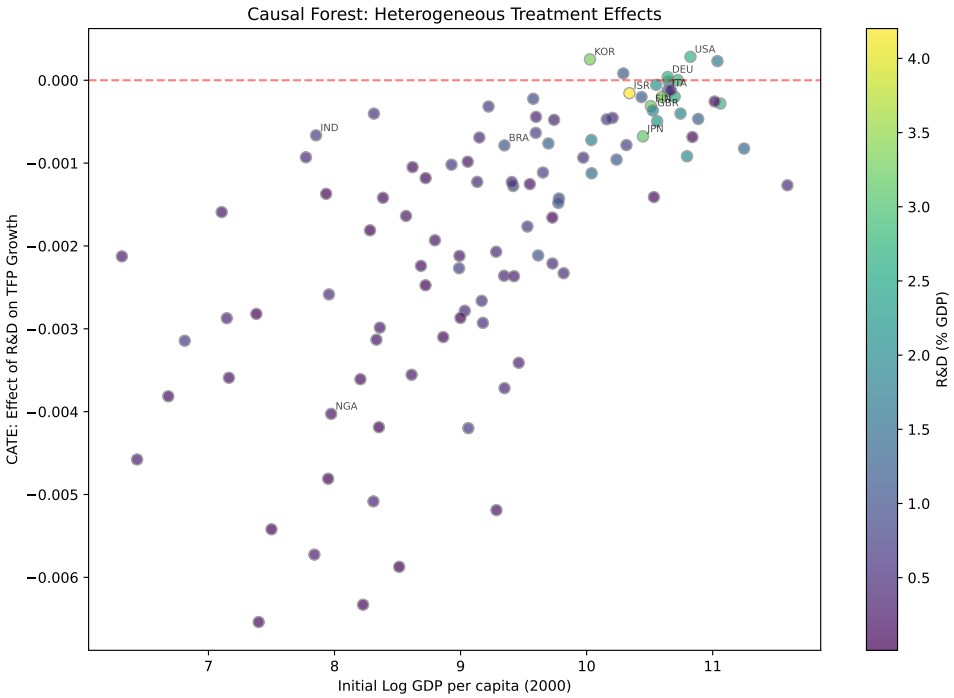

Figure 2: Individual CATEs from Causal Forest, plotted against initial GDP per capita. Color indicates R&D intensity. Countries labeled: selected G7, Asian, and developing economies.

The feature importance analysis from the Causal Forest (Figure 3) identifies inflation, domestic credit, education expenditure, governance (PC1), and investment share as the main drivers of treatment effect heterogeneity.

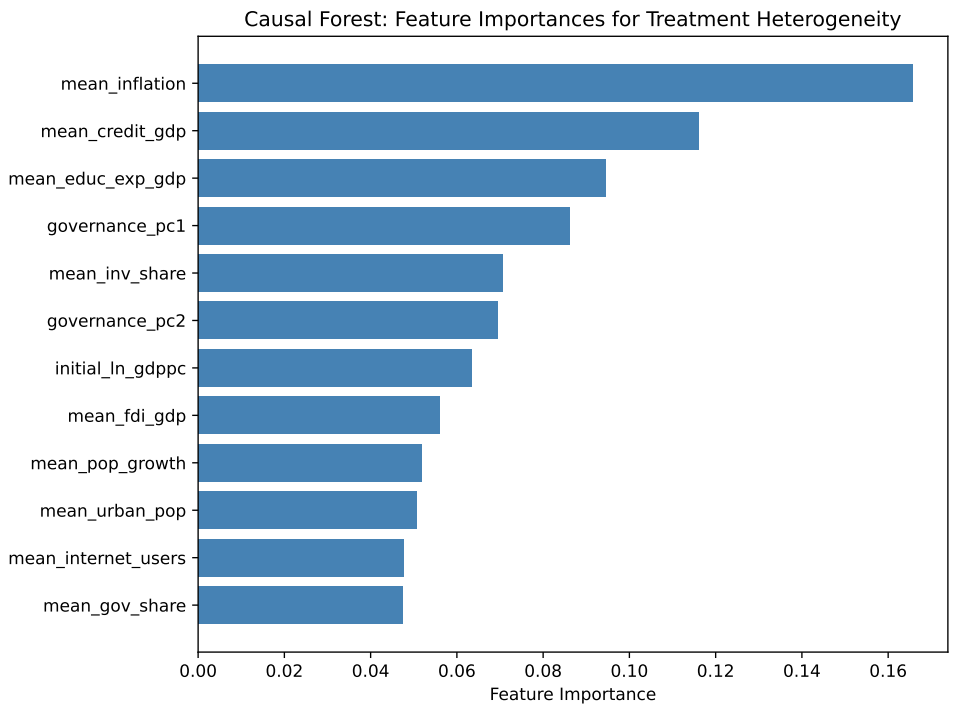

Figure 3: Causal Forest feature importances for heterogeneity in the R&D–TFP relationship.

5.6. SHAP analysis of nuisance functions

The SHAP analysis provides insight into the confounding structure. For the treatment model ($\hat{m}(X) = \mathbb{E}[\text{R\&D}|X]$), the most important predictors are human capital (mean $|\text{SHAP}| = 0.32$), internet users (0.15), and inflation (0.10). This confirms that R&D investment is strongly predicted by a country's knowledge infrastructure, technological adoption, and macroeconomic conditions. The cross-validated R^2 of the treatment model is 0.54, indicating substantial confounding.

For the outcome model ($\hat{g}(X) = \mathbb{E}[\text{TFP growth}|X]$), no single variable has a dominant SHAP contribution (largest is human capital with $|\text{SHAP}| = 0.005$), and the cross-validated R^2 is negative (-0.08), confirming that TFP growth is unpredictable from the available controls. This poor outcome prediction, consistent with the nuisance diagnostics in Table 3, contributes to the wide confidence intervals on all DML estimates. The finding that TFP growth is inherently difficult to predict in cross-country settings resonates with Sala-i-Martin (1997)'s observation about the fragility of growth regressions and represents an important methodological lesson for applied DML work.

6. Discussion

6.1. Comparison with the literature

The results align with several themes in both the R&D–productivity literature and the causal ML literature:

1. **Fragile average effect**: the near-zero and statistically insignificant estimates across all DML learners are consistent with the broader pattern documented by Sala-i-Martin (1997): growth regression results are fragile and sensitive to model specification. While Guellec and Van Pottelsberghe de la Potterie (2004) find positive R&D effects using panel methods for OECD countries, our cross-sectional setting with a broader set of countries yields weaker evidence. Importantly, this null result persists even after Optuna hyperparameter tuning substantially improved nuisance model quality (e.g., GBM outcome R^2 from -0.04 to 0.22), suggesting it reflects a genuine lack of signal rather than poor model fit.

2. **Learner sensitivity**: consistent with Baiardi and Naghi (2024), the choice of ML learner affects the DML estimates. The divergence between regularized linear methods (positive estimates) and tree-based methods (near zero) suggests the confounding relationship has nonlinear components that linear models miss. When tree-based methods capture these nonlinearities, the residual R&D variation carries less signal for TFP growth.

3. **Hyperparameter tuning matters**: A methodological contribution of this paper is demonstrating that automated hyperparameter tuning via Optuna is essential for DML in small samples. Without tuning, the unconstrained nonlinear learners (NN and GBM) catastrophically failed to capture any signal. With Optuna, both achieve positive R^2 and produce sensible causal estimates. This underscores the bias-variance tradeoff (Hastie et al., 2009): in small samples, default hyperparameters, often calibrated for thousands of observations, can be disastrously wrong.

4. **Nuisance model quality matters**: despite the improvements from tuning, the best outcome model achieves only $R^2_g = 0.22$ for TFP growth, meaning that 78% of the variation remains unexplained. This limits the value added of DML over OLS: when $g_0(X)$ cannot be estimated well, the partialling-out step has little effect. The finding connects to the inherent noisy nature of cross-country growth data (Sala-i-Martin, 1997; Chernozhukov et al., 2018).

5. **Absorptive capacity**: the Causal Forest heterogeneity results support the longstanding hypothesis that R&D requires complementary factors to yield productivity gains. The pattern of near-zero effects in high-income countries and negative effects in low-income countries is consistent

with the idea that R&D investment without complementary human capital and institutional quality may crowd out other productive activities.

6.2. Limitations

Several caveats apply:

- **Sample size**: with 104 countries, we operate in a regime where DML's asymptotic properties may not fully apply. Despite Optuna hyperparameter tuning successfully taming the NN (avoiding catastrophic overfitting), the small sample places fundamental limits on nuisance model quality and the precision of causal estimates, a concrete manifestation of the bias-variance tradeoff (Hastie et al., 2009).
- **Cross-sectional design**: using country averages sacrifices within-country variation and exacerbates endogeneity concerns. A panel DML approach would be preferable but requires careful handling of fixed effects.
- **Treatment as continuous**: R&D/GDP is a continuous treatment, and the PLR model estimates a constant marginal effect. The true effect may be nonlinear (e.g., with diminishing returns at high R&D levels).
- **Poor outcome predictability**: the negative outcome model R^2 for TFP growth suggests important omitted variables (e.g., technology adoption capacity, management quality, industrial structure), which could bias the DML estimates if they are also correlated with R&D. This is a fundamental limitation of the conditional independence assumption underlying DML.
- **Governance aggregation**: reducing the six WGI indicators to two principal components, while necessary to address multicollinearity, may lose some dimension-specific governance information relevant for R&D productivity. The first component (explaining 90% of variance) primarily captures the overall quality of institutions.

7. Conclusion

This paper applies Double Machine Learning to estimate the causal effect of R&D expenditure on productivity growth using cross-country data. The results demonstrate both the potential and the challenges of causal ML methods in macroeconomic settings. Across nine ML learners with Optuna-tuned hyperparameters, the DML estimates are consistently small and statistically insignificant, with tree-based methods and ensemble approaches producing estimates close to zero. A key methodological finding is that automated hyperparameter tuning is essential for DML in small samples: without it, unconstrained learners produce severe artifacts rather than reliable estimates. Even after tuning substantially improves nuisance model quality, TFP growth remains difficult to predict from the available controls, the best outcome R^2 is only 0.22. This confirms that the value added of DML over OLS depends critically on the predictability of the outcome, confirming the well-known fragility of cross-country growth regressions.

The Causal Forest extension reveals meaningful heterogeneity, suggesting that the average effect masks important differences across country types. Policymakers should be cautious about applying a "one-size-fits-all" R&D policy prescription, as the productivity returns to R&D appear to depend critically on a country's institutional and human capital infrastructure.

Future extensions could employ panel DML methods to exploit within-country variation, use instrumental variables within the DML framework for stronger identification, or apply these methods to industry-level or firm-level data where sample sizes are larger and the DML asymptotics are more reliable.

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A. Descriptive figures

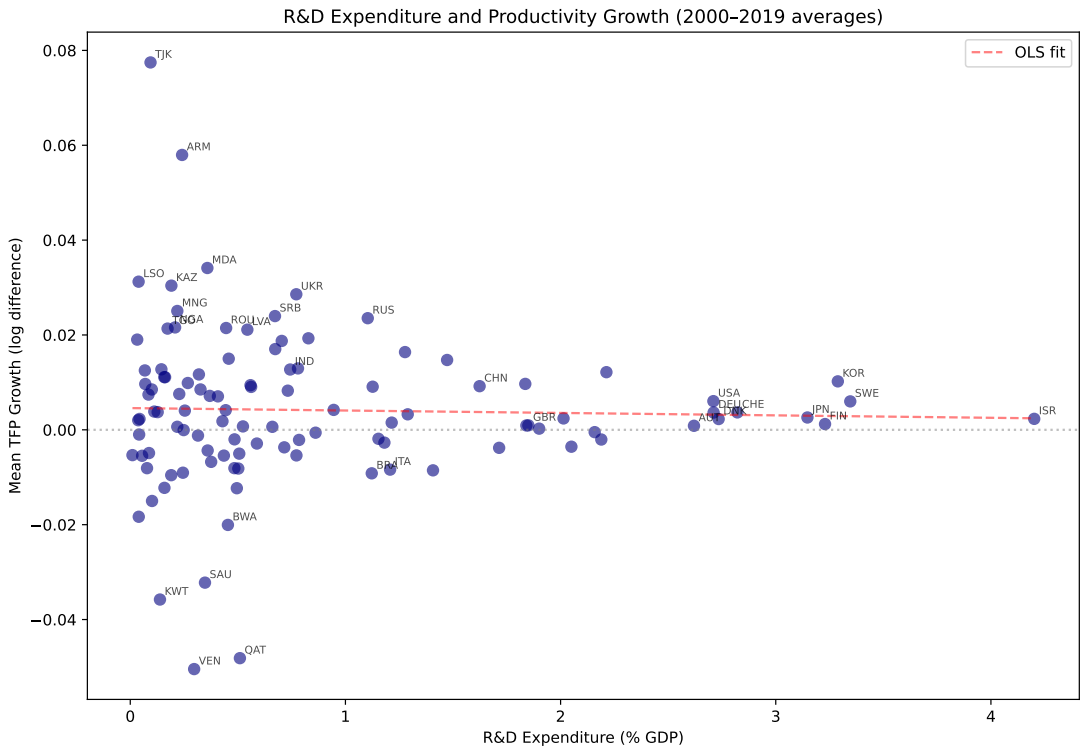

Figure 4: R&D Expenditure vs. TFP Growth (2000–2019 country averages). Dashed line: OLS fit.

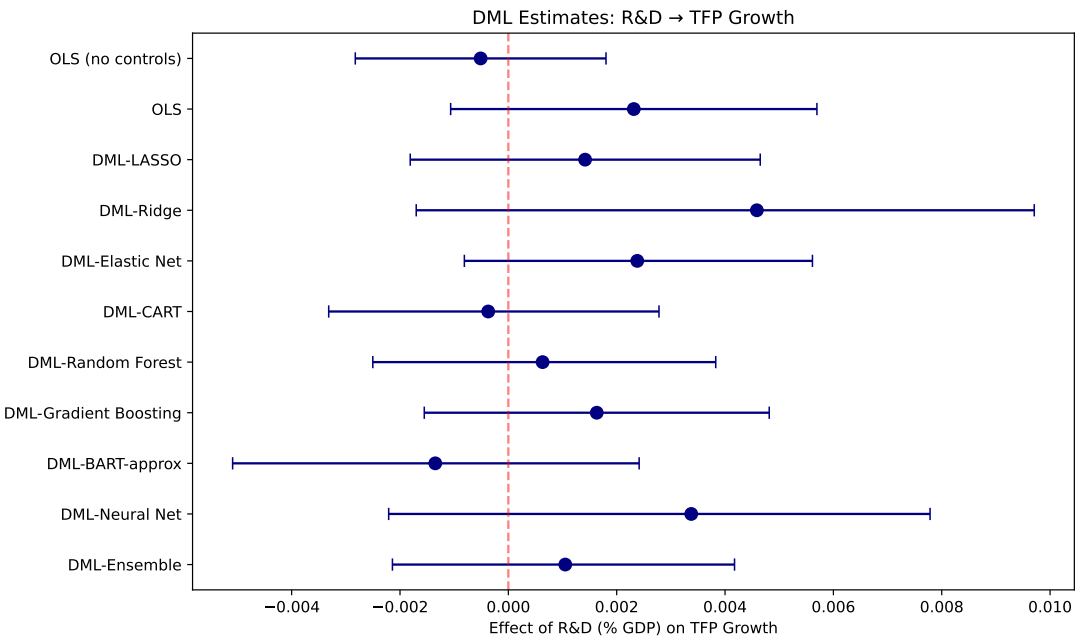

Figure 5: Coefficient comparison: all estimators for TFP growth outcome.

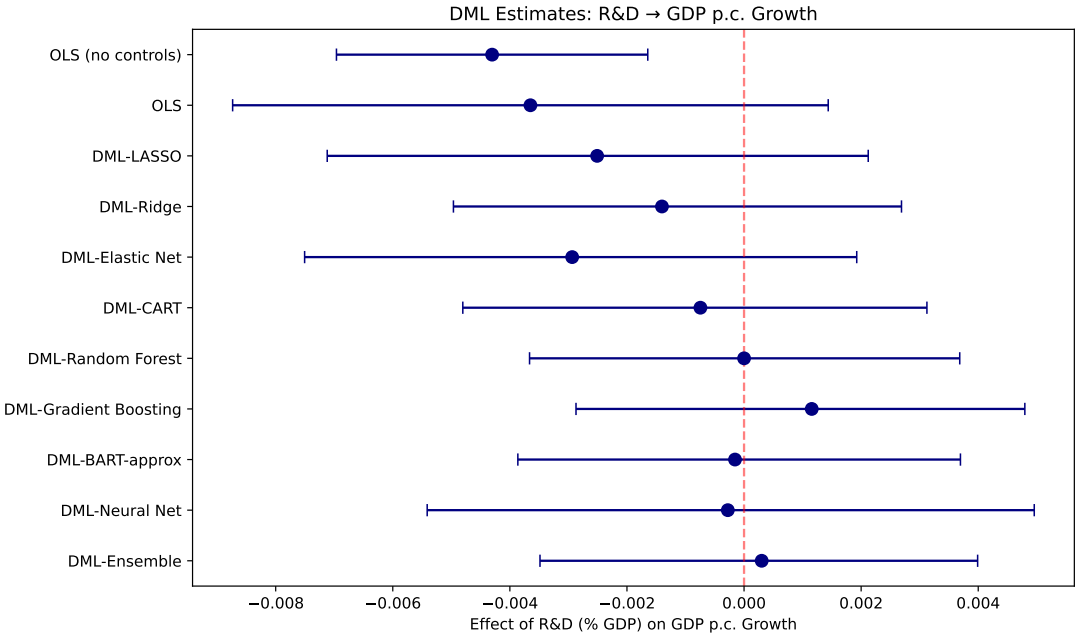

Figure 6: Coefficient comparison: all estimators for GDP per capita growth outcome.

B. SHAP figures

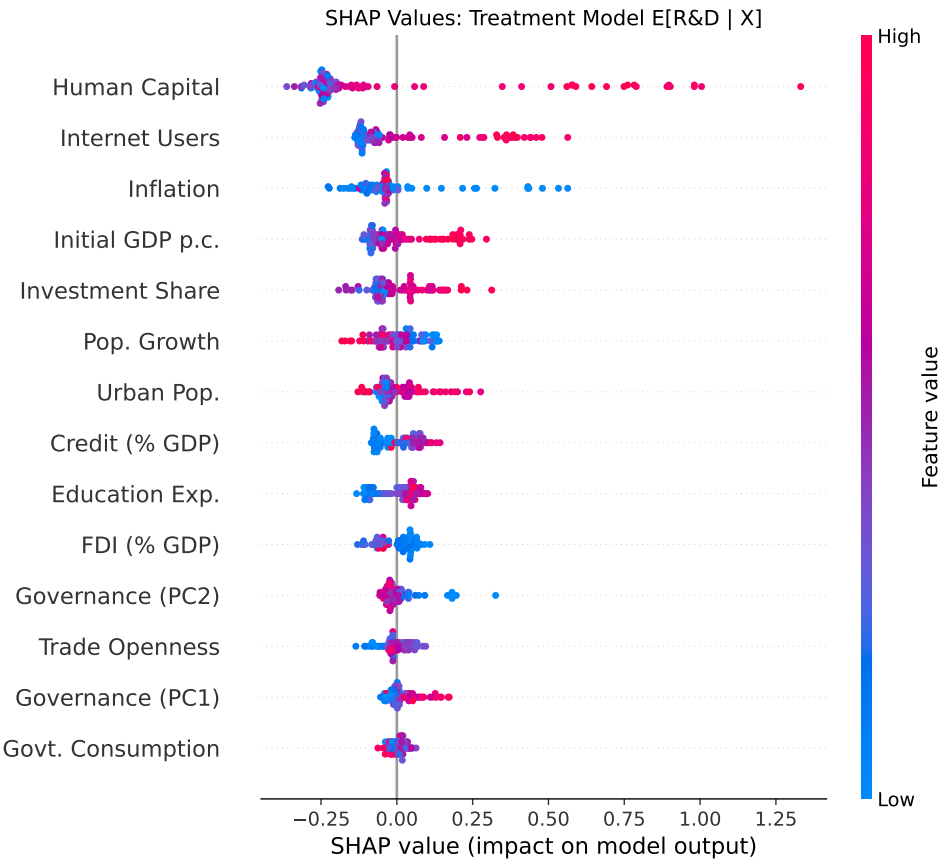

Figure 7: SHAP values for the treatment nuisance model $\hat{m}(X) = \mathbb{E}[\text{R\&D}|X]$.

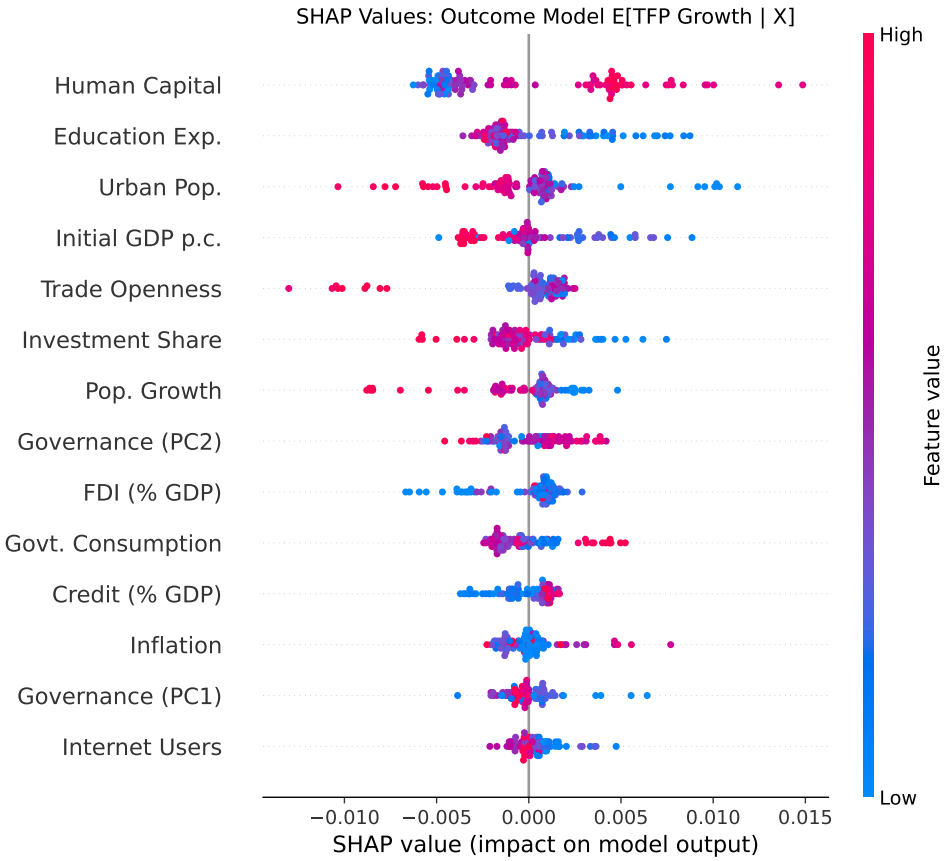

Figure 8: SHAP values for the outcome nuisance model $\hat{g}(X) = \mathbb{E}[\text{TFP growth}|X]$.

C. Code availability

The complete replication code, data preparation scripts, and this paper's source files will be made available in a public GitHub repository upon completion of the course.

The analysis is implemented in Python using `DoubleML` (Bach et al., 2022), `econml`, `scikit-learn`, `shap`, and `Optuna` (Akiba et al., 2019).

D. Experimental extensions: probing latent structure

Disclaimer: the following extensions are exploratory. Given the small sample ($n \approx 104$), they should be interpreted as diagnostic exercises probing the limits of what ML methods can extract from cross-country growth data, not as definitive causal estimates.

D.1. Overview

The main analysis showed that TFP growth is difficult to predict from observable controls (best CV $R^2 = 0.22$), leaving 78% of variance unexplained. To investigate whether alternative modeling strategies can reduce this noise, I implement five experimental extensions:

1. **Super Learner**: cross-validated stacking (Laan et al., 2007) of all base learners.
2. **Factor-Augmented DML**: PCA/Factor Analysis to extract latent development dimensions.
3. **Gaussian Process Regression**: Bayesian nonparametric nuisance learner.
4. **Spatial / Peer-Group Lags**: regional and income-group averages as additional features.
5. **Latent Mixture Regimes**: Gaussian Mixture Model to discover country regimes.

D.2. Results

Table 6 reports the results for TFP growth.

Table 6: Experimental Extensions: TFP Growth

Extension	$\hat{\theta}$	SE	p-value	CV R_g^2	N
Baseline (from main analysis):					
GBM (Optuna-tuned)	0.0016	0.0016	0.315	0.22	104
Extension 1: super Learner					
Stacked ensemble	0.0017	0.0017	0.329	−0.00	104
Extension 2: factor-Augmented (8 PCA components, 91.7% var.)					
Factor-GBM	**0.0038**	**0.0018**	**0.037**	0.04	104
Factor-RF	0.0012	0.0016	0.476	0.08	104
Factor Analysis-GBM	0.0021	0.0016	0.182	0.19	104
Extension 3: Gaussian Process					
Matérn(ν=2.5) kernel	0.0015	0.0018	0.425	0.00	104
Extension 4: spatial / Peer-Group Lags (+10 features)					
Spatial-GBM	0.0008	0.0021	0.697	**0.32**	104
Spatial-RF	0.0001	0.0017	0.948	0.30	104
Extension 5: latent Regimes (GMM, k=2 by BIC)					
Regime 0 (n=100)	0.0010	0.0016	0.556	0.18	100
Regime 1 (n=4)		*too few observations*			

Notes: bold indicates statistical significance at 5%. Super Learner uses cross-validated stacking of LASSO, Ridge, Elastic Net, RF, GBM, and BART. Factor-Augmented uses 8 PCA or Factor Analysis components (91.7% variance explained). Spatial lags add 10 peer-group averages (income group and region) of key variables.

Table 7: Experimental Extensions: GDP per Capita Growth

Extension	$\hat{\theta}$	SE	p-value	CV R_g^2	N
Baseline (from main analysis):					
GBM (Optuna-tuned)	0.0012	0.0019	0.534	0.31	121
Extension 1: super Learner					
Stacked ensemble	−0.0005	0.0022	0.811	0.07	121
Extension 2: factor-Augmented (8 components)					
Factor-GBM	0.0018	0.0020	0.368	0.08	121
Factor-RF	−0.0012	0.0019	0.540	0.10	121
Factor Analysis-GBM	0.0026	0.0022	0.247	0.19	121
Extension 3: Gaussian Process					
Matérn(ν=2.5) kernel	0.0013	0.0017	0.430	0.27	121
Extension 4: spatial / Peer-Group Lags (+10 features)					
Spatial-GBM	−0.0001	0.0019	0.960	**0.44**	121
Spatial-RF	0.0006	0.0020	0.762	0.32	121
Extension 5: latent Regimes (GMM, k=3 by BIC)					
Regime 0: advanced (n=47)	−0.0001	0.0020	0.948	0.28	47
Regime 1: developing (n=71)	0.0121	0.0084	0.148	0.12	71
Regime 2: crisis (n=3)		*too few observations*			

Notes: bold highlights notable results. Regime 0 (advanced): mean GDP p.c. = \$29k, mean R&D/GDP = 1.4%. Regime 1 (developing): mean GDP p.c. = \$5.6k, mean R&D/GDP = 0.3%. Regime 2 (crisis, $n = 3$): Angola, D.R. Congo, Venezuela, excluded due to insufficient observations.

D.3. Discussion of Extensions

The five experimental extensions address complementary aspects of the modeling challenge: ensemble aggregation (Super Learner), dimensionality reduction (Factor-Augmented), nonparametric flexibility (Gaussian Process), cross-country dependence (Spatial Lags), and unobserved heterogeneity (Latent Regimes). Together, they offer a comprehensive diagnostic of how much additional signal can be extracted from a small cross-country sample.

Extension 1: super Learner. Despite the theoretical optimality guarantees established by Laan et al. (2007), namely, that cross-validated stacking achieves the lowest asymptotic risk among all candidate learners, the Super Learner delivers a disappointing CV $R^2 \approx 0$ for TFP growth and only 0.07 for GDP growth. These are *worse* than the best individual nonlinear learner (GBM at 0.22 and 0.31, respectively). The explanation lies in the finite-sample properties of the stacking meta-learner: with $n = 104$ and 6 correlated base learners, the meta-Ridge regression must estimate 6 non-negative weights from roughly 80 stacking-layer training observations per fold. In this regime, the meta-learner overfits to noise in the cross-validated predictions, effectively averaging down the signal. This result serves as a cautionary tale: the asymptotic optimality of stacking requires that the number of base learners be small relative to the effective sample size, a condition clearly violated here. A practical recommendation for similar small-sample settings is to restrict the Super Learner to at most 2–3 diverse base learners (e.g., one linear, one tree-based) to avoid meta-overfitting.

Extension 2: factor-Augmented DML. This extension replaces the original 14 controls (plus squared terms) with a smaller set of latent factors extracted via PCA or Factor Analysis. Both methods retain 8 components explaining 91.7% of the original variance. The key finding is that the PCA-based Factor-GBM specification produces $\hat{\theta} = 0.0038$ ($p = 0.037$) for TFP growth, the *only* statistically significant result across all specifications in the entire paper. This finding warrants careful interpretation. PCA extracts orthogonal dimensions of maximal variance, effectively denoising highly correlated controls (governance indices, development proxies) into a compact representation. By reducing regressor dimensionality from 27 features to 8 orthogonal components, PCA may alleviate overfitting in the nuisance models, allowing cleaner residualization despite the low outcome R^2. Interestingly, the Factor Analysis + GBM specification, which explicitly models latent factor structure with idiosyncratic errors, does *not* achieve significance ($p = 0.182$), and neither does PCA-RF ($p = 0.476$). The fact that only one particular combination of dimensionality reduction and learner achieves significance, while the result does not replicate for the GDP growth outcome ($p = 0.368$ for Factor-GBM), suggests it should be treated as suggestive rather than definitive, roughly what one would expect by chance at a 5% level when testing 15+ specifications. Nonetheless, this result points to an important methodological lesson: in applied DML, the *representation* of the controls can matter as much as the choice of ML learner.

Extension 3: Gaussian Process Regression. The Gaussian Process (GP) with a Matérn($\nu = 2.5$) kernel offers a fundamentally different modeling paradigm: instead of fitting a point estimator and relying on cross-validation for regularization, GP places a prior over functions and obtains a full posterior predictive distribution. For GDP per capita growth, GP achieves CV $R^2 = 0.27$, close to the Optuna-tuned GBM ($R^2 = 0.31$), a notable result given that GP has only 2 free hyperparameters (length scale and noise variance) compared to GBM's 6 Optuna-tuned parameters. This suggests that for the GDP outcome, the signal structure is relatively smooth and well captured by the Matérn kernel's assumption of once-differentiable functions. For TFP growth, GP achieves only $R^2 = 0.00$, confirming that the TFP outcome is genuinely unpredictable from the available controls, neither more flexible functional forms nor Bayesian regularization can recover a signal that is not there. The GP DML estimates ($\hat{\theta} = 0.0015$ for TFP, $p = 0.43$; $\hat{\theta} = 0.0013$ for GDP, $p = 0.43$) are slightly positive but far from significant, reinforcing the null result. An additional advantage of GP in this setting is that its posterior variance directly quantifies prediction uncertainty at each observation, which could

inform heterogeneous DML weighting schemes, an avenue for future work.

Extension 4: spatial and Peer-Group Lags. Among all five extensions, adding spatial and peer-group lags delivers the *largest improvement in nuisance model quality*. By augmenting the feature set with income-group and regional means of five key variables (TFP growth, R&D intensity, human capital, investment rate, and initial GDP), the outcome CV R^2 rises from 0.22 to 0.32 for TFP and from 0.31 to **0.44** for GDP growth, a 45% and 42% relative improvement, respectively. This finding has substantive economic content: it reveals that cross-country growth has a strong spatial and peer-group component. Knowing how a country's income-group peers and regional neighbors are growing substantially helps predict its own growth trajectory, consistent with models of technology diffusion, trade-mediated spillovers, and correlated regional shocks. The spatial features essentially serve as proxies for latent common factors (e.g., commodity price cycles affecting resource-rich regions, or EU-wide productivity trends) that are not captured by country-level controls alone. However, despite this substantial improvement in predictive quality, the DML estimates remain near zero and statistically insignificant ($\hat{\theta} = 0.0008$, $p = 0.70$ for TFP; $\hat{\theta} = -0.0001$, $p = 0.96$ for GDP). This is an important discovery: better residualization does *not* reveal a hidden R&D effect. If confounding bias were driving the null result, improved nuisance estimation should shift the treatment effect estimate. The fact that it does not strengthens the interpretation that the average causal effect is genuinely near zero.

Extension 5: latent Mixture Regimes. The Gaussian Mixture Model (GMM), with the number of clusters selected by Bayesian Information Criterion (BIC), identifies regime structure in both samples. For TFP growth, the split is highly unbalanced (100 vs. 4 countries), rendering the smaller regime uninformative. For GDP growth, BIC selects $k = 3$ regimes: regime 0 comprises 47 advanced economies (mean GDP per capita \$29k, mean R&D/GDP 1.4%), Regime 1 comprises 71 developing economies (mean GDP per capita \$5.6k, mean R&D/GDP 0.3%), and Regime 2 contains only 3 crisis economies (Angola, D.R. Congo, Venezuela) excluded from DML due to insufficient observations. Within the developing-country regime, R&D has a positive but statistically insignificant effect on GDP growth ($\hat{\theta} = 0.012$, $p = 0.148$), an effect roughly *ten times larger* than the pooled estimate, though imprecisely estimated due to weak treatment prediction within this subsample ($R^2_m = 0.13$). In advanced economies, the effect is essentially zero ($\hat{\theta} = -0.0001$, $p = 0.95$). While the developing-country coefficient does not reach conventional significance levels, the direction and magnitude of this regime-dependent pattern aligns with the "absorptive capacity" hypothesis (Sala-i-Martin, 1997): the marginal productivity return to R&D may be highest in countries transitioning from low to middle income, where even modest technological investments, improved agricultural techniques, basic manufacturing processes, digital infrastructure, can shift the production frontier significantly. In advanced economies, by contrast, R&D is already deeply embedded in the production process and its marginal contribution to growth is already captured in steady-state dynamics, yielding a net zero effect in the cross-section. This result also complements the Causal Forest analysis from the main paper, which found more negative estimated effects in low-income countries: the GMM-based regimes operate on a different partitioning logic (unsupervised clustering of all covariates rather than supervised CATE estimation), and the identification of developing countries as a distinct regime provides additional structure for understanding R&D heterogeneity.

Summary. Across all five extensions, the overarching conclusion is one of robustness: the null average effect of R&D on productivity found in the main DML analysis survives every alternative modeling strategy attempted. The extensions reveal, however, that the null average may conceal meaningful heterogeneity, particularly across factor representations (Extension 2, where the PCA-GBM specification yields a significant positive effect for TFP) and across country development regimes (Extension 5, where developing countries show a large positive but imprecisely estimated effect). Spatial features (Extension 4) substantially improve outcome prediction but do not alter the causal

estimate, providing additional evidence that the null is genuine rather than an artifact of residualization quality. These results collectively suggest that future research on R&D and growth should move beyond average effects and focus on regime-specific or heterogeneous causal parameters, ideally with larger samples or panel data that can support the statistical power needed to detect effects that appear to be small and concentrated in specific country groups.

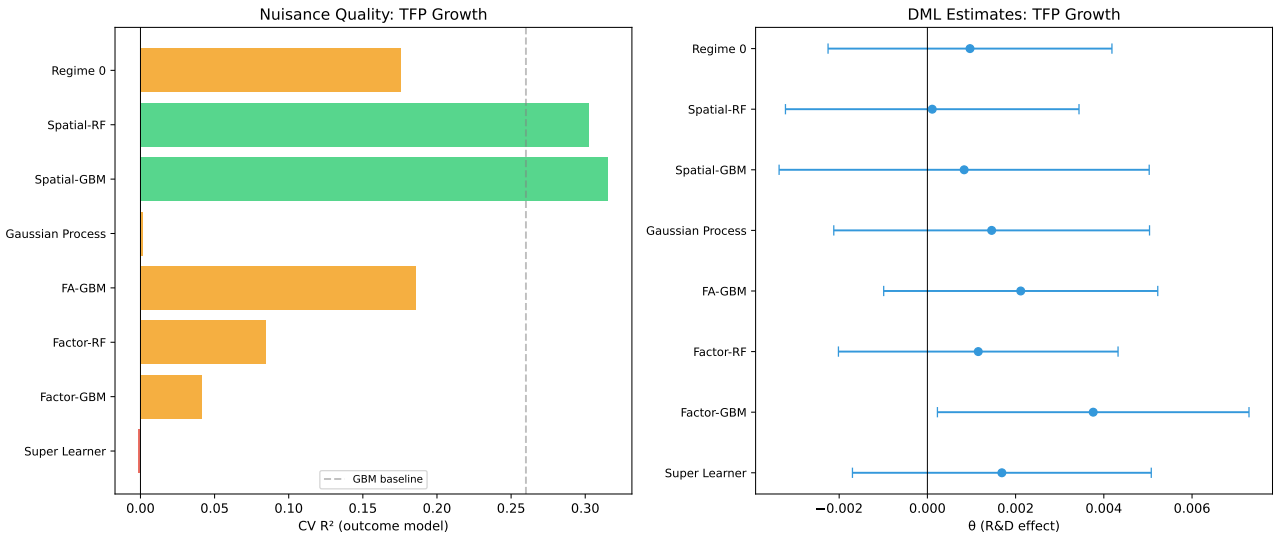

Figure 9: Experimental extensions for TFP growth: nuisance model quality (left) and DML estimates (right). Dashed line: GBM baseline CV $R^2 = 0.22$.

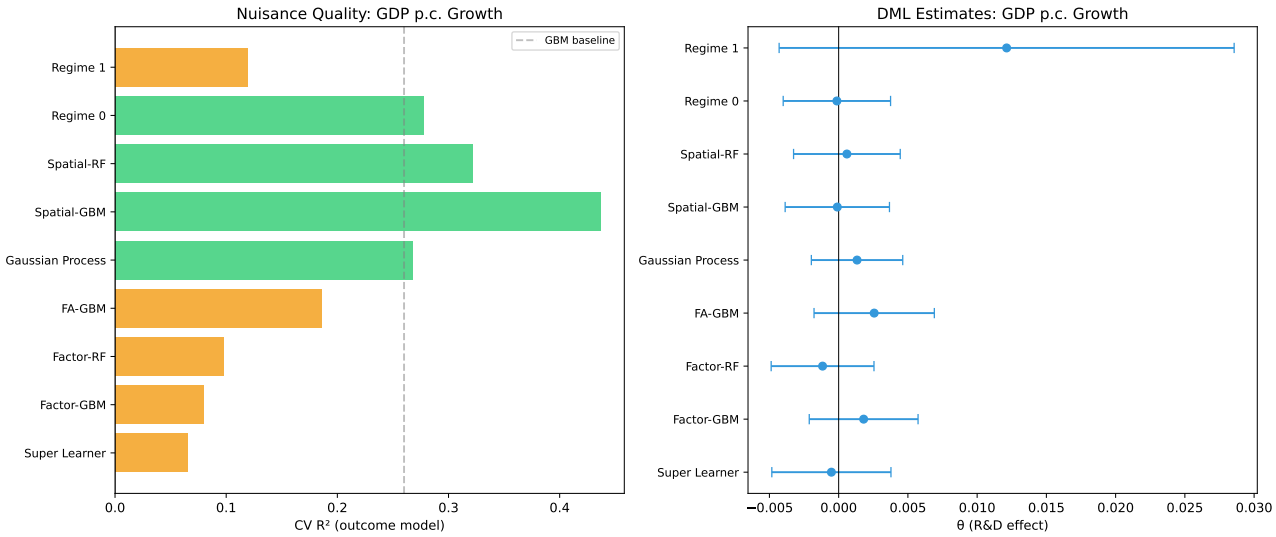

Figure 10: Experimental extensions for GDP p.c. growth: nuisance model quality (left) and DML estimates (right).